

The $\cos 2\phi$ modulation on tensor-polarized ${}^6\text{Li}$: projected inclusive and coherent measurements

2026-08-17 · development run 5 · polligen (evgen/) + polli_fastsim (fastsim/) · companions: plans/06 (background budget), docs/note_cos2phi_coherent_6Li.md, reports/polarized_li_primer.html (pedagogical introduction) · external numbers verified against primary sources; this revision adds dedicated fact checks of the moment constraint and of the far-forward detection chain

This note presents two projected EIC measurements of the $\cos 2\phi$ azimuthal modulation on a transversely tensor-polarized ${}^6\text{Li}$ beam: inclusive DIS, requiring only the scattered electron, and coherent diffraction with the ${}^6\text{Li}$ detected intact in its 1^+ ground state. The modulation arises from photon double helicity flip and probes gluon transversity, to which neither quarks nor bound nucleons contribute at leading twist [1,2]; it has not been measured on any target. The projections are given together with the assumptions entering each number and the checks performed on them.

13.5%

coherent-tag acceptance at IP6 with the 0.20 GeV envelope cut, $\exp(-B \cdot p_T^2 / \text{cut})$, $B = 50 \text{ GeV}^{-2}$ [9–20% for $B \in 40\text{--}60$]. The cut is documented for top-rigidity beams; for ${}^6\text{Li}$ it is an optics assumption (§4.1).

$1.1 \times 10^7 / 10^8$

tagged coherent events in 1 / 10 EIC years (scenario $f_0 = 0.04$); best-bin $\delta A = 1.9 \times 10^{-3} / 6 \times 10^{-4}$, i.e. 5σ floors at 1.0% / 0.3% modulation

0.036

deformation-anchored (a_2) of the tagged sample at $P_{zz} = 0.6$ [band 0.018–0.059; one-sided $\times 0.73$ rate-weighting systematic], scaled from the polarized-deuteron CGC calculation [13]; sign flip vs the deuteron predicted

1 • The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

The $\cos 2\phi$ term requires the virtual photon to flip its helicity by two units. Since no spin-1/2 constituent can absorb two units of helicity, neither quarks nor bound nucleons contribute at leading twist: $\Delta(x, Q^2)$ is purely gluonic and vanishes for a nucleus composed of nucleons alone [1]. At leading order it is generated by the gluon transversity distribution δG through the $O(\alpha_s)$ photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left(\sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

(in the sign convention of Ref. [25]; conventions differ between papers). Δ has not been measured for any target; the only prior experimental document is a JLab letter of intent without a sensitivity projection [3]. Quantitative expectations are scarce. The only moment estimate is the Sather–Schmidt bag-model calculation for the spin-3/2 Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [4], which suggests asymmetries at the per-mille level; percent-level values arise only in maximal-transversity ($\Delta_T g = \Delta g$) scenarios for proton–deuteron Drell–Yan, described by their authors as optimistic [5]. The projections therefore span the scenario band $\Delta/F_1 \in 10^{-3}\text{--}10^{-2}$.

The measured amplitude and its inversion back to the structure function are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = \hat{A} \frac{y^2 D_\phi}{1-y}$$

The fastsim approximation set is $A_{//} = D \cdot g_1 / F_1$ (the y^2 -suppressed term dropped), $g_2 = g_2^{\text{WW}}$, and no target-mass corrections. The last matters only at high x and low Q^2 ; the sweet-spot bins sit at $x \leq 0.06$, where $y^2 = 4M^2 x^2 / Q^2 < 10^{-2}$, so it is a labeling caveat rather than a bias. Structure functions are toy parameterizations with

PDF-grid (CT18NLO/NNPDFpol11) backends behind the same interface: rates are reliable to a factor of ≈ 1.5 , and asymmetry shapes are scenarios by construction.

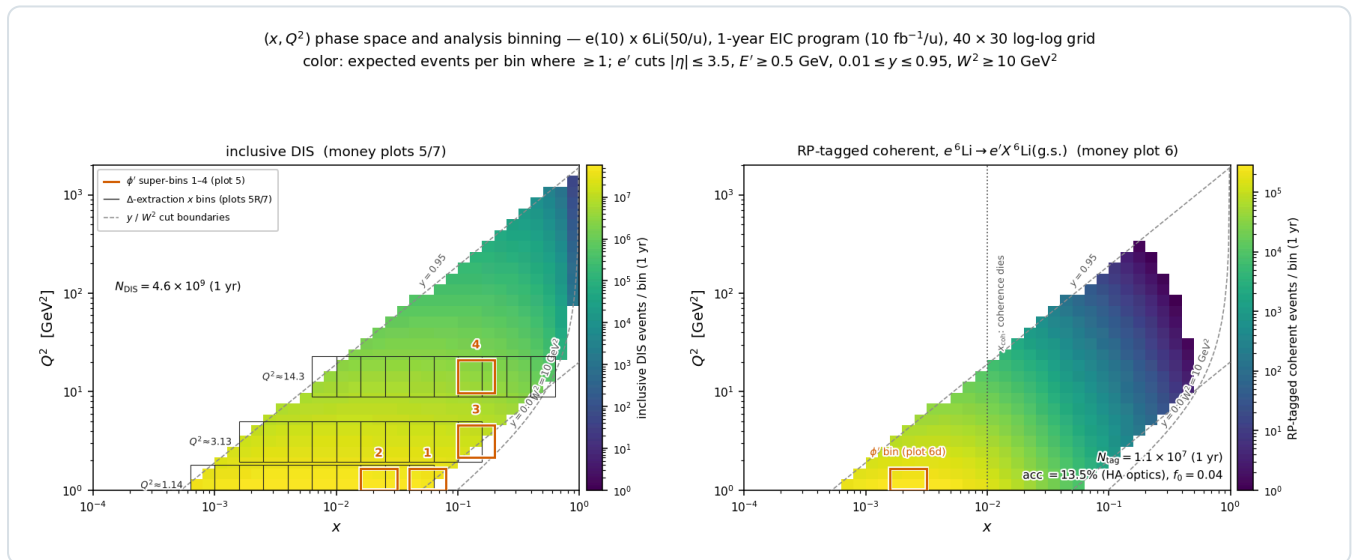
2 • Machinery, statistics, and binning

Generator. No public generator produces polarized e+A events for $A > 1$ (BeAGLE is unpolarized; DJANGO and PEPSI are nucleon-level and longitudinal-only). polligen therefore implements only the missing component, the spin-density matrix $\otimes$ structure-function master formula with spin-correlated sampling, and relies on existing tools for the rest. Its validation gates are bin-wise identity with the analytic asymmetry formulas at relative tolerance 10^{-12} , pseudo-experiment spreads that close on every analytic error map, and unbiased $\cos 2\phi$ recovery under ϕ -acceptance holes.

Full-luminosity statistics. For a binned estimator, drawing each ϕ -bin count from a Poisson distribution with the exact expected yield is statistically identical to event-level sampling, so a full-luminosity pseudo-experiment costs 24 Poisson draws (`sample.phi_histogram_pseudo`). Event-level sampling remains available and is what the closure tests use.

Estimator. The amplitude comes from a binned least-squares fit of $N(\phi') = C(1 + A \cos 2\phi')$, not the $2\langle \cos 2\phi' \rangle$ moment: with realistic ϕ holes the moment is biased by more than the signal, while the fit stays unbiased. Errors follow $\delta A = \sqrt{(2/N)/P_{ZZ}}$, validated by pseudo-experiments. The fit carries no $\cos \phi'$ term; the m-symmetric tensor fills set $a_1 = 0$ by construction.

Binning. The four displayed (x, Q^2) super-bins are selected by statistical significance within Q^2 bands: an unconstrained ranking concentrates at the lowest Q^2 , while the Q^2 lever arm provides the consistency check against power-suppressed contributions (§6). Each super-bin merges 3×3 cells of the 40×30 analysis grid. Rates use per-nucleon luminosity, F_2^A/A , scattered-electron cuts $|\eta| \leq 3.5$, $E' \geq 0.5$ GeV, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$ GeV², and the full luminosity in the transverse-tensor fill; projections are quoted for the 1-year ($10 \text{ fb}^{-1}/u$) and 10-year ($100 \text{ fb}^{-1}/u$) programs.



Phase-space coverage and analysis binning ($e^{10} \times {}^6\text{Li}50$, 1-year program, 40×30 log-log grid; color: expected events per bin where ≥ 1 ; dashed lines: y and W^2 cut boundaries). **Left, inclusive DIS** (4.6×10^9 accepted events in year 1): boxes 1–4 are the ϕ' super-bins of money plot 5; gray combs are the three Q^2 slices ($Q^2 \times [1/1.6, 1.6]$) whose merged x-bin pairs form the data points of the amplitude-versus- x panel and of the Δ extraction (plots 5R/7), retained where $N \geq 10^3$ and $\delta A \leq 8 \times 10^{-3}$. **Right, RP-tagged coherent channel:** the same map multiplied by $f_{\text{coh}}(x)$ and the 13.5% tag acceptance. $N_{\text{tag}} = 1.1 \times 10^7$ in year 1; 98% of the tagged sample lies at $x < 0.03$ (median 3.5×10^{-3}). The box is the super-bin of money plot 6(d); plot 6(c) is this map summed over Q^2 .

3 • The Δ model and the moment constraint

Provenance. Sather and Schmidt computed the lowest moment of Δ in the MIT bag model for the spin-3/2 Δ^{++} baryon, together with its Q^2 evolution; they provide no x dependence [4]. The explicit $\alpha_s(Q^2)$ is the leading-

order Wilson coefficient of the photon–gluon box, so the $O(\alpha_s)$ suppression relative to F_1 is model-independent; the value -0.012 is the bag estimate. Lattice QCD provides two related results, each with substantial caveats: the bare ϕ -meson matrix element $A_2 = 0.23(2)(5)$ at $m_\pi = 450$ MeV [25] converts to $\int x \Delta|_\phi \approx -0.006 \alpha_s$ (our conversion, unrenormalized), while the NPLQCD deuteron calculation gives a δG moment of $-0.010(3)$, bare at $m_\pi \approx 806$ MeV [27], a transversity-to-momentum ratio roughly ten times below that of the ϕ and implying the opposite sign for the deuteron's Δ moment. No calculation of Δ exists for ${}^6\text{Li}$ or any $A > 2$ nucleus; the only nuclear model is a $\Delta\Delta$ -isobar estimate for the deuteron [34]. The literature thus brackets the nuclear Δ in both directions: suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect, as argued in the polarized-ion white paper [50].

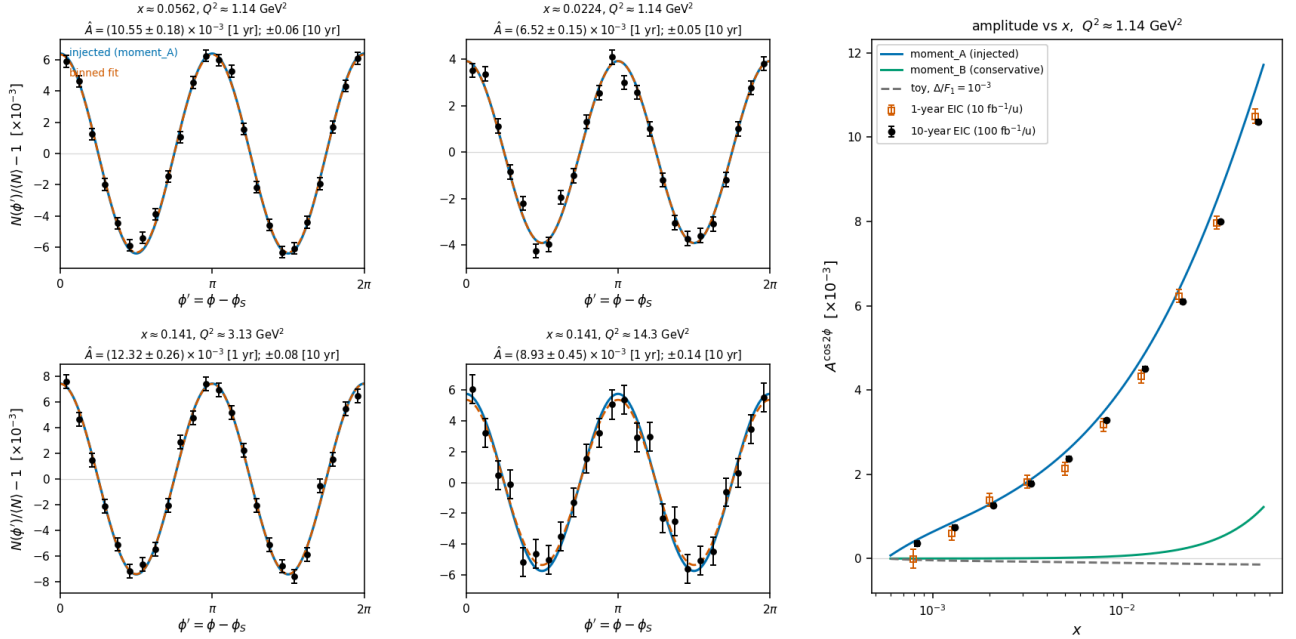
Implementation. All Δ models reside in one registry (`polli_fastsim/delta_models.py`). The production default, `moment_A` , is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ${}^6\text{Li}$:

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.294$$

with $(a, b) = (0.7, 3)$; the factor $1/3$ restricts the exotic glue to the polarized deuteron-like pair, while the whole-nucleus P_{ZZ} remains a separate factor, so that no double counting occurs. The constraint is imposed at the rate-weighted $\langle Q^2 \rangle = 3.9$ GeV² of the accepted phase space; because F_1 evolves, the realized moment varies between -4% and $+7\%$ across the plotted Q^2 slices. α_s is taken from the CT18NLO table (the leading-order analytic fallback is 14% higher). The solved $A = -0.294$ is consistent with the EPPS21-based `money_delta` value of -0.31 at its $\langle Q^2 \rangle = 7.4$ GeV². `moment_B` applies the same constraint without the F_1 factor and yields amplitudes a factor of 2–8 lower; the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot.

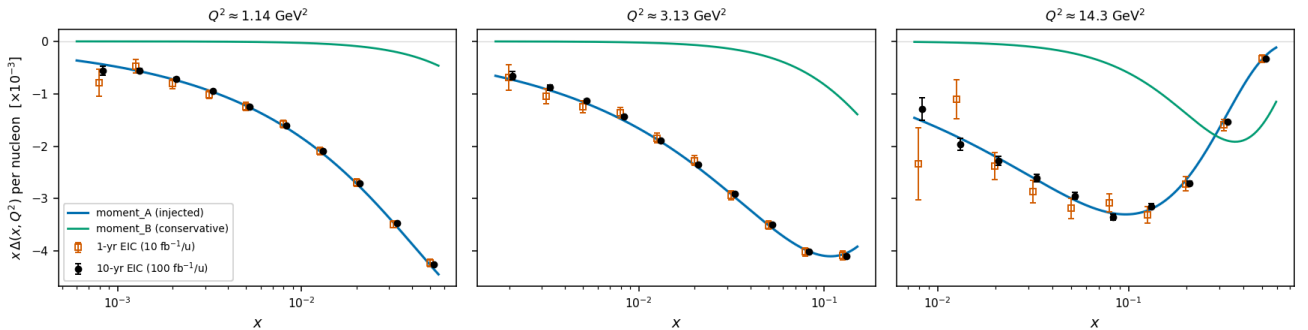
Status of the normalization. Transferring a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent in the literature; both the normalization and the x dependence are labeled scenario choices, bracketed by `moment_B` from below and by the possible nuclear enhancement from above. The projections demonstrate that the measurement has the precision to decide: under the moment constraint, the x dependence separates interpretations A and B bin by bin.

Nuclear gluonometry, transversely tensor-polarized ${}^6\text{Li}$, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
moment_A(variant=mid_x, c=-0.012, A=-0.2941, q2_ref=3.913, dilution=0.3333); ϕ' at 1 yr (10 fb^{-1}/u), amplitudes 1 & 10 yr — stat. only; bkg & tensor RC unquantified (plans/0)



Money plot 5 — inclusive gluonometry ($e10 \times {}^6\text{Li}50$, $P_{zz} = 0.60$, Δ : moment_A, dilution 1/3). Left 2×2 : ϕ' pseudo-data with 1-year statistical error bars in the four sweet-spot super-bins ($Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$, $x \approx 0.02\text{--}0.14$); injected curve (blue), binned fit (vermillion); annotations give δA for both programs. Right: extracted amplitude vs x with 1-year (open) and 10-year (filled) error bars against the moment_A, moment_B, and toy- 10^{-3} curves. Amplitudes are $(0.7\text{--}1.2) \times 10^{-2}$ at the sweet spots against $\delta A = (1.5\text{--}4.5) \times 10^{-4}$ in year 1, i.e. every bin $\geq 21\sigma$, so the A-versus-B spread is resolved by the x dependence. Only the scattered electron is detected.

Extracted double-helicity-flip $\Delta(x, Q^2)$ of ${}^6\text{Li}$, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
 $\hat{\Delta} = \hat{A} y^2 D_{\phi}/(1-y)$ per bin (model bin-centering; dilution 1/3 incl.; stat. only, bkg & tensor RC unquantified); area = S-S moment $-0.012 \alpha_s/3$



Money plot 7 — the extracted $\Delta(x, Q^2)$. The amplitudes of money plot 5 unfolded to the structure function (model bin-centering, $\delta\Delta = \delta A \cdot |\partial\Delta/\partial A|$), shown as $x\Delta$ per nucleon at the three sweet-spot Q^2 slices with independent 1-year (open) and 10-year (filled) pseudo-measurements against the moment_A and moment_B curves; the area under each curve is the ported bag moment ($\div 3$ dilution). Best bins reach $\delta\Delta \approx (0.7\text{--}1.4) \times 10^{-3}$ in year 1 on $\Delta \approx -0.03\text{--}-0.09$.

4 • The coherent channel

4.1 • Detecting the intact ${}^6\text{Li}$ at ePIC

In coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, the recoil retains the beam's magnetic rigidity ($A/Z = 2$; the diffractive momentum loss $x_p < 0.01$ changes it by less than 1%) and emerges at $\theta = p_T/p \lesssim 1.7 \text{ mrad}$. Charged particles below 5 mrad enter the far-forward system: the Bo spectrometer (5.5–20 mrad, charged tracks and photons above $\sim 50 \text{ MeV}$), the Off-Momentum Detectors ($\sim 30\text{--}60\%$ of beam rigidity), the Roman Pots at $z \approx 26/28 \text{ m}$ ($\theta < 5 \text{ mrad}$, $\sim 60\text{--}95\%$ rigidity for off-rigidity tracks, AC-LGAD sensors with $\sim 140 \mu\text{m}$ spatial and $\sim 30 \text{ ps}$ timing resolution), and the ZDC for neutrals [7,8,9]. A recoil at beam rigidity remains inside the beam

envelope and is visible only in the Roman Pots, and only above the 10σ envelope cut; at low $|t|$ the intact recoil cannot be tagged at IP6 [10,51]. With an exponential coherent t -distribution the tag acceptance is analytic:

$$\varepsilon_{\text{tag}} = e^{-B p_{T,\text{cut}}^2} = 13.5\% \quad (p_{T,\text{cut}} = 0.20 \text{ GeV}, B = 50 \text{ GeV}^{-2}); \quad 4 \times 10^{-5} \quad (0.45 \text{ GeV})$$

Provenance of the cuts. The 0.20 GeV floor is the documented EIC requirement for high-acceptance (low-divergence) optics at top rigidity, corresponding to a beam divergence $\leq 73 \mu\text{rad}$ for 275 GeV protons [7,48], and matches the one documented light-ion number: coherent ^3He tagging limited to $p_T > 0.20 \text{ GeV}$ at IR-6 [10]. The 0.45 GeV value is our derivation (150 μrad high-divergence proton optics applied to the 300 GeV ^6Li momentum), not a documented number. No beam-optics parameters have been published for any lithium beam; if the ^6Li beam ran with proton-class divergence at this rigidity, the 10σ cut would fall at 0.5–0.7 GeV and eliminate the IP6 channel. This cut is the most consequential machine input to the measurement (assumption #1, §6).

Coherence selection at IP6 and at IR-8. Standard EIC practice at IP6 is to establish coherence by vetoing breakup rather than by tagging the recoil: dissociation protons (rigidity 0.50) reach the Off-Momentum Detectors, neutrons the ZDC, de-excitation photons the Bo calorimeter and ZDC, and ^3He -like fragments (0.75) the Roman-Pot window [51]. Direct low- $|t|$ tagging is a central argument for the second interaction region, whose lattice adds a secondary beam focus $\sim 45 \text{ m}$ downstream where the beam size is $\sigma \approx 0.15 \times 0.1 \text{ mm}$, allowing Roman Pots at $z = 44.0/45.5 \text{ m}$ to reach $p_T \approx 0$. The published intact-recoil efficiencies there are 47% (d), 32% (^3He), 29% (^4He), and 17.8% (^7Li) at top energy, rising steeply at lower energies (^3He reaches 99.8% at 5×41) [10]; no ^6Li case has been published, and our interpolation is $\approx 20\%$. One degeneracy remains even there: a Roman-Pot hit measures only A/Z , so intact ^6Li , α , and d at the same velocity are identical in trajectory and timing. The second-detector design addresses this with a proposed Z^2 -scaling Cherenkov counter behind the pots, currently at the R&D stage [8]. For the IP6 AC-LGADs no charge-discrimination study exists (signal amplitude is used only for position interpolation), and the minimum-angle ion acceptance there is not yet fully established [49]. This question, open item #19, determines whether IP6 coherent-tag purity is established event by event or statistically (§5).

4.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 \quad [0.02\text{--}0.08]$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the f_0 band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-nucleus saturation estimates (20–25% [45]). The scale $x_{\text{coh}} = 0.01$ marks where the coherence length $l_c = 0.105 \text{ fm}/x_p$ falls below the ^6Li size. The slope $B = 50 \text{ GeV}^{-2}$ [40–60] follows from $B = \langle r^2 \rangle / 3$ with the matter radius (the charge radius gives ≈ 57 [47]; gluonic distributions may be wider [42]); the exponential form is valid below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [11,12]. Parameterizing f_{coh} in Bjorken x rather than x_p is optimistic for high-mass diffraction, which the f_0 band is intended to cover. The resulting yields are 1.1×10^7 tagged events in year 1 and 1.1×10^8 over ten years, with best-super-bin $\delta A = 1.9 \times 10^{-3}$ and 6×10^{-4} respectively.

4.3 • The anchored amplitude

The $\cos 2\phi'$ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen, and Zhao [13], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [14,15]). Their coefficients, digitized in `coherent.MANTYSAARI_A2_DEUTERON`, show a_2 growing linearly in $|t|$ with $a_2(m=0) = -2a_2(m=\pm 1)$, arising from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz} :

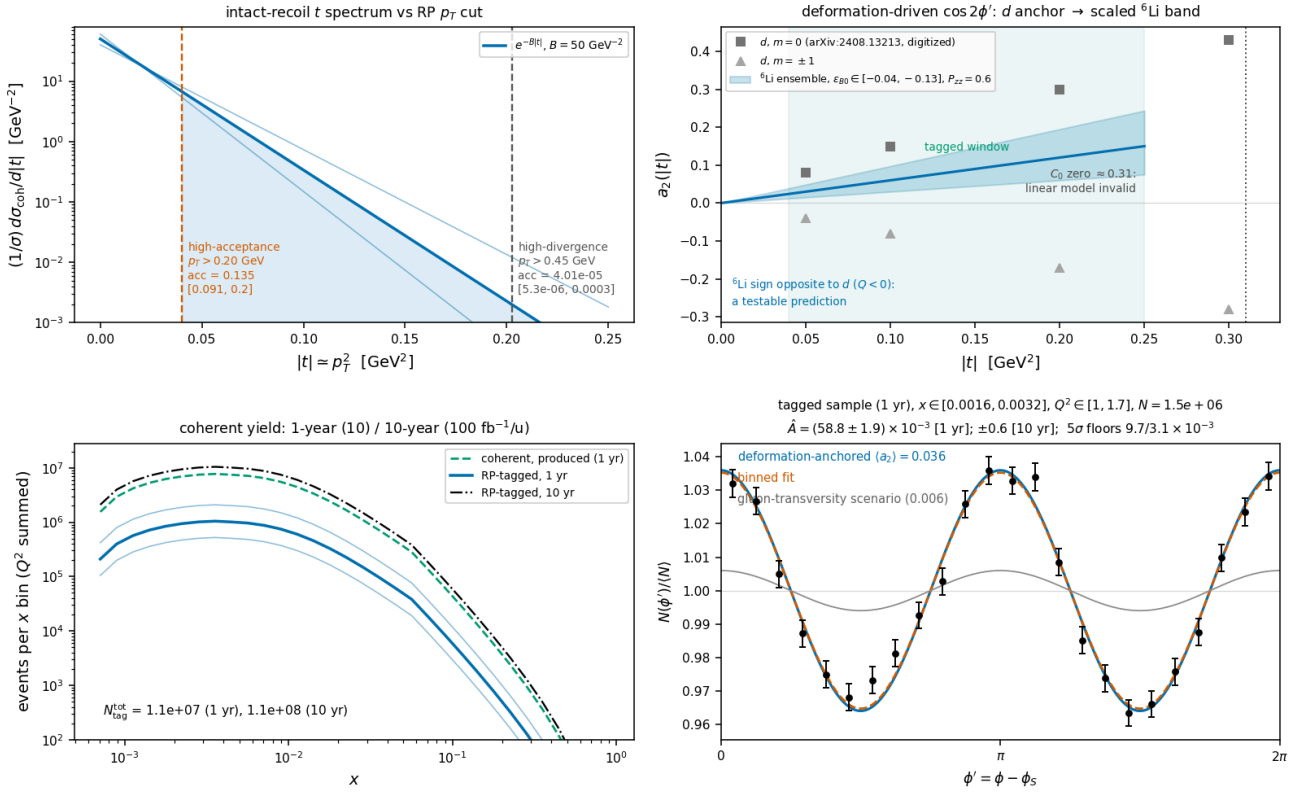
$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \quad [0.018\text{--}0.059]$$

Convention note (added 2026-08-24). The anchor's Eq. (9) expands $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$, with Φ the azimuth of the produced vector meson — the momentum transfer, i.e. the recoil — relative to the polarization axis in the γ^*d frame. The $\cos 2\Phi$ modulation coefficient is therefore $2a_2$, and the $\langle a_2 \rangle$ quoted here and injected in money plot 6(d) understates the deformation modulation by a factor two (0.072 rather than 0.036 at $P_{zz} = 0.6$); this report's coherent numbers are conservative by $\times 2$ and are kept in the a_2 convention until the WP5 curves are redone. The modulation is one of the *recoil* azimuth $\varphi_t - \varphi_S$, not of the electron azimuth used as the axis of panel (d); the reconstructed-level companion (reconstruction-chain report §7, money plot 6R) injects the corrected coefficient in the recoil azimuth and fits both azimuths.

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron value of 0.21 using two indicators that disagree: the relative quadrupole deformation $Q/\langle r^2 \rangle$ of ${}^6\text{Li}$ is about one fifth of the deuteron's and opposite in sign, while the α -d asymptotic D/S ratio is disputed (a transfer measurement gives $\eta = +0.0003(9)$, consistent with zero [16], whereas older sub-Coulomb analyses favored $\eta \approx -0.01$). We adopt $\varepsilon_{B0} \in -(0.04-0.13)$ with default -0.08 , leaving near-zero deformation open; the 5σ floors cover its detection or exclusion. The width of the band reflects genuine theoretical uncertainty: even exact few-body calculations miss the ${}^6\text{Li}$ quadrupole cancellation by a factor of ≈ 3 [17], and the gluonic deformation need not share the cancellation observed in the charge sector. The sign flip relative to the deuteron is a testable prediction ($Q({}^6\text{Li}) < 0$). Tensor modulations of comparable size are established at these $|t|$: e-d $T_{20} = -(0.1-0.3)$ [19] and HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [20], the latter attributed to coherent double scattering [21–23]. The flat gluon-transversity term is separately bounded at $3 \times 10^{-3} - 10^{-2}$, from the bare lattice ϕ transversity [25] divided by the $\sim 10\times$ NPLQCD deuteron suppression [27] (both at unphysical pion masses), with the maximal Drell–Yan scenario [5] as the ceiling. The equal-rate population average behind $\langle a_2 \rangle$ carries a one-sided $\times 0.73$ rate-weighting systematic (audited; within the ε_{B0} band).

Separation of the two mechanisms. The mechanisms differ in shape: deformation is linear in t , sign-flipped relative to the deuteron, and expected to weaken toward small x_P (JIMWLK evolution washes out deformation [15]); exotic glue is flat in t and persists at small x , where it connects to the elliptic-gluon program [6]. In a two-component fit $a_2(t) = c_{\text{def}}|t| + a_g$ per x_P bin, both components exceed the statistical floor by $\geq 5\sigma$ at $100 \text{ fb}^{-1}/u$. Incoherent breakup is m -state-blind in this mechanism [13]: it dilutes the modulation but cannot mimic it.

Coherent $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$, transversely tensor-polarized, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
 nario rates; modulation anchored on the polarized- d CGC calculation, PLB 858:139053 -- plans/06 SS6.4b; statistical only, backgrounds/purity and tensor RC unquanti



Money plot 6 — coherent channel, $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$ (same beam, $P_{zz} = 0.60$). Top left: recoil t -spectrum vs both optics' p_T cuts with the slope band. Top right: digitized deuteron a_2 per m -state [13] vs the scaled ^6Li ensemble band (sign flip predicted); tagged window and form-factor-zero exclusion marked. Bottom left: tagged yield vs x for both programs with the f_0 band ($N_{\text{tag}} \approx 1.1 \times 10^7 / 1.1 \times 10^8$). Bottom right: ϕ' pseudo-data (1-year statistics) at the anchored $\langle a_2 \rangle = 0.036$ with the flat gluon-transversity scenario (gray); extraction $\hat{A} \pm 1.9 \times 10^{-3}$ (1 yr), $\pm 0.6 \times 10^{-3}$ (10 yr).

5 • Backgrounds for the coherent tag (full budget: plans/06)

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, $\Gamma = 24$ keV, decays $\sim 100\%$ $\alpha + d$ [39])	Both fragments at $R = 1.000$, same trajectory and velocity as intact ^6Li : tracking and ToF blind; only dE/dx ($\propto Z^2$: 9/4/1) separates	RP charge discrimination — open question #19 : no study exists for the IP6 AC-LGADs (amplitude is used for position; the documented Z-ID is the proposed IR-8 Cherenkov [8]). Fallback: two-component $ t $ fit (coherent $e^{-50} t $ vs $\sim 10\times$ flatter incoherent; crossover at 0.05–0.2 GeV 2 inside the window [41,42]); $e + \text{Pb}$ benchmark: 80–99% incoherent rejection [43]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [39]	A fragment could pass unnoticed	$p \rightarrow \text{OMD}$ ($R = 0.50$), $n \rightarrow \text{ZDC}$, $^3\text{He} \rightarrow \text{RP window}$ ($R = 0.75$) [51]: vetoable; inefficiency is second-order (BeAGLE study to quantify)
$^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2$ eV) [39]	Particle-stable ($\alpha + d$ parity-forbidden, $\Gamma_{\alpha} \leq 6 \times 10^{-7}$ eV); γ -decays to a real $A = 6$, $Z = 3$ track	Isoscalar exchange cannot drive $T = 0 \rightarrow 1$; the boosted 0.1–0.4 GeV γ lands mostly in the B0 calorimeter (photons ≥ 50 MeV, 5.5–20 mrad [9]), tail in the ZDC. The 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is γ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t -dependence = charge FF^2 [11,12]); subtract and reuse as an RP-acceptance candle; the charge-quadrupole $\cos 2\phi$ is $< 10^{-3}$ for $ t < 0.15$ GeV 2 [12,17]
Beam-gas / halo; e' -side fakes	Accidental $R \approx 1$ tracks; π / pair-symmetric e' at high y	Non-colliding bunches, ~ 30 ps timing, vertex association; standard DIS e' ID plus (e' , recoil) kinematic consistency

Concerning extraction systematics: the modulation is self-normalized within a single fill, so relative-luminosity systematics cancel; ϕ -acceptance holes are handled by the binned fit; the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$ is common with A_{zz} ; and radiative corrections for tensor observables have not been calculated, which is noted on every plot. At the level of interpretation, a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [34] and conventional double scattering generates the low- x b_1 [21–23]; the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison discriminate against both.

The ${}^6\text{Li}$ null test. Deformation-driven coherent modulations, the polarized-deuteron imaging signal [13], are bounded on ${}^6\text{Li}$ by its anomalously small quadrupole moment, $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [35]. A sizable flat- $\text{in-}t \cos 2\phi$ on ${}^6\text{Li}$ therefore cannot be a shape effect; the mechanisms that are entangled for the deuteron separate in the ${}^6\text{Li}/d$ comparison.

6 • Critical assumptions

#	ASSUMPTION	IMPACT IF WRONG	REPLACEMENT PATH
1	RP p_T cuts: 0.20 GeV is the documented top-rigidity envelope scale (${}^3\text{He}$ precedent [10]); 0.45 GeV is our derivation; no Li optics exist — proton-class divergence would give 0.5–0.7 GeV	Tag acceptance anywhere from 13.5% to negligible at IP6; the physics case shifts to IR-8	Li beam-optics parameters from EIC C-AD; the IR-8 secondary focus removes the cut entirely [10]
2	Scenario coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$], parameterized in Bjorken x	Coherent yields scale linearly; δA as $1/\sqrt{f_0}$	Real diffractive model — open question #18; concrete ask: an IP-Glasma α -d run [13]
3	Bag-moment normalization of Δ ported to per-nucleon ${}^6\text{Li}$ with 1/3 dilution; x -shape ours	Overall Δ scale; literature brackets both directions (NPLQCD suppression [27] vs binding enhancement [50])	Lattice moment for a light nucleus; theory engagement (plans/04)
4	ϵ_{B0} scaling from the deuteron (two dials disagreeing by $\times 3$)	Deformation amplitude anywhere in the band; sign flip robust	The same IP-Glasma α -d calculation; band quoted on every plot
5	Event-by-event Z-ID in the RPs is NOT assumed (purity via $ t $ -shape fit)	If Z-ID exists purity improves; if the incoherent slope model is off, the fit systematic grows	#19 answer from the ePIC far-forward group; BeAGLE incoherent spectra
6	Linear $a_2(t)$ valid only for $ t < 0.2 \text{ GeV}^2$; form-factor zero at 0.31 GeV^2 excluded, not modeled	Upper tagged window carries unmodeled structure	Form-factor-level model of the dip region (deuteron analog in [13])
7	TOY structure functions; no target-mass terms; $g_2 = g_2^{WW}$	Rates to $\times 1.5$; shapes are scenarios	PDF-grid backends exist; digitized theory curves scheduled (plans/02)
8	All luminosity in the transverse-tensor fill; $P_{zz} = 0.6$ in-ring placeholder	N scales with the run-plan share; amplitude with P_{zz}	Run-plan bookkeeper supports splits; source demonstrations reach $\sim 95\%$ of maximal P_{zz} [37]
9	Impulse approximation; no FSI; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; RC unquantified	Theory engagement (plans/04); the QED tail is calculable and doubles as calibration
10	Equal-rate population average for $\langle a_2 \rangle$; t_{min} neglected in the tag acceptance	One-sided $\sim 27\%$ (rate weighting) and $\sim 10\%$ (t_{min}) — inside the ϵ_{B0} / f_0 bands	Rate-weighted ensemble model (the IP-Glasma run supersedes both)

7 • Next steps

Near term: BeAGLE incoherent ${}^6\text{Li}$ spectra to quantify the veto efficiency and the $|t|$ -fit systematics (plans/02); VMC α -d/ α -t overlaps to replace the two-parameter radial forms (plans/04); pseudo-experiment closure with a two-dimensional (ϕ , ϕ_t) acceptance map (plans/03); and HepMC3 output with an ePIC chain test (plans/05), so that these projections can be repeated at reconstructed level.

Theory: a rerun of the IP-Glasma polarized-deuteron setup [13] with an α -d cluster density, the highest-leverage input, which would replace both ϵ_{B0} indicators and the f_0 band at once; a lattice Δ moment for a light

nucleus; the spin-3/2 structure-function basis for ${}^7\text{Li}$; tensor-observable radiative corrections; and the sum-rule connection between transversity-GPD moments and measured tensor form factors [30].

Machine and detector: lithium beam-optics parameters, which set the IP6 p_T cut (assumption #1); an answer to open item #19, whether the RP AC-LGAD amplitude can separate $Z = 3, 2, 1$ at identical rigidity and velocity; and the IR-8 secondary focus with its proposed Z^2 Cherenkov [8], which extends the tag toward $p_T \approx 0$ (${}^7\text{Li}$: 17.8% published; ${}^6\text{Li}$ not yet studied [10]). The coherent- ${}^6\text{Li}$ program adds one more physics case for the second interaction region. Off-momentum halo determines how closely the pots approach the beam, and therefore the cut itself.

References

- [1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.
- [2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.
- [3] J. Maxwell et al., *Search for Exotic Gluonic States in the Nucleus*, JLab LOI12-16-006, arXiv:1803.11206.
- [4] E. Sather, C. Schmidt, *Size and scaling of the double-helicity-flip hadronic structure function*, Phys. Rev. D 42 (1990) 1424.
- [5] S. Kumano, Q.-T. Song, Phys. Rev. D 101 (2020) 054011, arXiv:1910.12523; Phys. Rev. D 101 (2020) 094013, arXiv:2003.06623.
- [6] Y. Hatta, B.-W. Xiao, F. Yuan, Phys. Rev. Lett. 116 (2016) 202301.
- [7] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419.
- [8] Far-forward/second-IR design: ATHENA collaboration, JINST 17 (2022) P10019, arXiv:2210.09048; *Precision Studies of QCD in the Low Energy Domain of the EIC*, arXiv:2211.15746 (SVIII); BNL LDRD 23-050 closeout (second-detector conceptual design, Mini-DIRC Z^2 Cherenkov), arXiv:2602.04636.
- [9] M. Pitt, PoS DIS2024 (2025) 259, arXiv:2409.02811; E.-C. Aschenauer et al., Phys. Rev. D 112 (2025) 036010, arXiv:2503.05908.
- [10] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, *Opportunities for imaging light nuclei with a second interaction region at the EIC*, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638.
- [11] L. R. Suelzle, M. R. Yearian, H. Crannell, Phys. Rev. 162 (1967) 992.
- [12] G. C. Li, I. Sick, R. R. Whitney, M. R. Yearian, Nucl. Phys. A 162 (1971) 583.
- [13] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213.
- [14] H. Mäntysaari, H. Roch, B. Schenke, C. Shen, W. Zhao, Phys. Rev. D 114 (2026) 014068, arXiv:2605.00454.
- [15] H. Mäntysaari, P. Singh, Eur. Phys. J. C 85 (2025) 1449, arXiv:2411.14934.
- [16] K. D. Veal, C. R. Brune, W. H. Geist et al., Phys. Rev. C 60 (1999) 064003.
- [17] R. B. Wiringa, R. Schiavilla, Phys. Rev. Lett. 81 (1998) 4317, nucl-th/9807037.
- [19] M. Bouwheuis et al. (NIKHEF), Phys. Rev. Lett. 82 (1999) 3755; D. Abbott et al. (JLab t20), Phys. Rev. Lett. 84 (2000) 5053; D. M. Nikolenko et al. (VEPP-3), Phys. Rev. Lett. 90 (2003) 072501.
- [20] A. Airapetian et al. (HERMES), Phys. Rev. Lett. 95 (2005) 242001, hep-ex/0506018.
- [21] N. N. Nikolaev, W. Schäfer, Phys. Lett. B 398 (1997) 245.
- [22] K. Bora, R. L. Jaffe, Phys. Rev. D 57 (1998) 6906, hep-ph/9711323.
- [23] J. Edelmann, G. Piller, W. Weise, Phys. Rev. C 57 (1998) 3392.
- [25] W. Detmold, P. E. Shanahan, Phys. Rev. D 94 (2016) 014507 [Erratum: Phys. Rev. D 95 (2017) 079902], arXiv:1606.04505.
- [27] F. Winter et al. (NPLQCD), Phys. Rev. D 96 (2017) 094512, arXiv:1709.00395.
- [30] W. Cosyn, B. Pire, Phys. Rev. D 98 (2018) 074020, arXiv:1806.01177; W. Cosyn et al., Phys. Rev. D 99 (2019) 094035.
- [34] M. Nzar, P. Hoodbhoy, *Estimation of the double helicity flip deuteron structure function*, Phys. Rev. D 45 (1992) 2264.
- [35] I. Angeli, K. P. Marinova, At. Data Nucl. Data Tables 99 (2013) 69; N. J. Stone, At. Data Nucl. Data Tables 111–112 (2016) 1.
- [37] M. M. Dalton, A. Deur, C. D. Keith, Eur. Phys. J. A 61 (2025) 111, arXiv:2504.21177.
- [39] D. R. Tilley et al., *Energy levels of light nuclei A = 5, 6, 7*, Nucl. Phys. A 708 (2002) 3 (TUNL).
- [41] LEPS collaboration, coherent ϕ on ${}^4\text{He}$, arXiv:1711.01095.
- [42] STAR collaboration, Phys. Rev. Lett. 128 (2022) 122303, arXiv:2109.07625.
- [43] W. Chang et al., Phys. Rev. D 104 (2021) 114030, arXiv:2108.01694.
- [44] H1 collaboration, Eur. Phys. J. C 72 (2012) 2074, arXiv:1203.4495.
- [45] T. Lappi, Nucl. Phys. A 830 (2009) 403c, arXiv:0907.4588; H. Kowalski, T. Lappi, C. Marquet, R. Venugopalan, Phys. Rev. C 78 (2008) 045201, arXiv:0805.4071.
- [47] I. Sick, arXiv:1505.06924.
- [48] ECCE consortium, exclusive/diffractive/tagging studies, NIM A 1052 (2023) 168238, arXiv:2208.14575.
- [49] J. R. Pybus, S. Li, L. Calero-Diaz, *Polarized nuclear DVCS at the EIC* ($e+{}^3\text{He}$), arXiv:2606.11491.
- [50] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794.
- [51] IR-8 far-forward tagging of nuclear breakup, arXiv:2501.12410.

checks of the Sather-Schmidt moment usage and the far-forward detection chain. Reference numbering is shared with earlier revisions; retired entries are omitted. Target venue: INT program on polarized ion beams at the EIC, March 22 – April 2, 2027.